ROYAL FITNESS CLUB

Fitness Foundations Certificate

MODULE 1

How the Body Works — Anatomy Essentials

Duration	45 minutes
Course Code	CS_B1
Level	Beginner
Format	Study Guide + Reference PDF

SECTION 1: Introduction to Human Anatomy

Why Anatomy Matters for Fitness

Before picking up a dumbbell or lacing up your running shoes, understanding the machine you are training transforms how effectively you exercise. Human anatomy is not just a medical subject — it is your personal operating manual. Knowing where each muscle is, how it attaches, and what movement it produces lets you train with precision rather than guesswork.

In this module we explore the foundational systems of the human body — skeletal, muscular, and connective tissue — with a practical fitness lens. By the end you will be able to identify major muscle groups, understand joint mechanics, and appreciate why proper form protects your body for decades of training.

The Organisation of the Human Body

The body is organised into levels of complexity: cells → tissues → organs → organ systems → organism. For fitness professionals, the three most critical organ systems are:

- Skeletal System — the rigid framework that provides structure and leverage
- Muscular System — the contractile engine that generates movement
- Nervous System — the control network that triggers and coordinates muscle contractions

Key Learning Outcome
• Identify the 206 bones of the human skeleton by region
• Distinguish between the axial and appendicular skeleton
• Understand how skeletal structure influences exercise selection

SECTION 2: The Skeletal System

Bones — Far More Than a Scaffold

The adult human skeleton contains 206 bones. At birth we have roughly 270 — many fuse during childhood and adolescence, a process completing around age 25. This is why resistance training programmes for teenagers must be carefully managed: growth plates (epiphyseal plates) are still cartilaginous and vulnerable to compressive injury.

Axial vs Appendicular Skeleton

The skeleton is divided into two functional regions:

AXIAL SKELETON (80 bones)	APPENDICULAR SKELETON (126 bones)
Skull (22)	Shoulder girdle (4)
Vertebral column (26)	Arms & forearms (6)
Rib cage (25)	Hands & wrists (54)
Hyoid bone (1)	Pelvic girdle (2)
	Legs & feet (60)

Bone Composition and Density

Bone is a living tissue composed of an organic matrix (mainly collagen) reinforced with mineral crystals of hydroxyapatite (calcium phosphate). This composite structure gives bone its unique property of being both flexible and hard.

Compact bone (cortical): Dense outer shell — 80% of skeletal mass. Provides mechanical strength and houses the medullary cavity (where red/yellow marrow is found).

Cancellous bone (trabecular): Spongy inner lattice at bone ends. Lighter but still incredibly strong — oriented along lines of mechanical stress. This architecture is why bone remodels in response to exercise loads.

Bone remodelling cycle: Osteoclasts resorb old bone; osteoblasts deposit new bone. Resistance training creates microstress that upregulates osteoblast activity, increasing bone mineral density (BMD) — a key defence against osteoporosis later in life.

Fitness Fact — Bone & Exercise
• Weight-bearing exercise increases BMD by 1–3% per year in untrained individuals
• Impact sports (basketball, gymnastics) produce highest BMD gains
• Swimming, while excellent for cardiovascular health, provides minimal BMD benefit
• The "mechanostat" theory: bone adapts to habitual loading thresholds

The Vertebral Column — Your Central Tower

The spine consists of 33 vertebrae grouped into 5 regions, each with distinct biomechanical properties relevant to exercise programming:

Region	Vertebrae	Normal Curve	Primary Movement

Cervical	7 (C1–C7)	Lordosis	Flexion/Extension/Rotation
Thoracic	12 (T1–T12)	Kyphosis	Limited — rib attachment
Lumbar	5 (L1–L5)	Lordosis	Flexion/Extension
Sacral	5 fused	Kyphosis	None (fused)
Coccygeal	4 fused	—	None (fused)

The natural S-curve of the spine (two lordoses + two kyphoses) distributes compressive loads evenly across intervertebral discs. When this neutral alignment is lost during lifting (e.g., excessive lumbar flexion during deadlifts), shear forces increase dramatically — a primary cause of disc herniation.

Intervertebral Discs

Between each pair of vertebrae (except C1–C2) sits an intervertebral disc — a fibrocartilaginous shock absorber. Each disc has:

- Nucleus pulposus — gelatinous core, 70–90% water, absorbs compressive load
- Annulus fibrosus — concentric collagen rings providing tensile strength
- Cartilaginous endplates — nutrient exchange interface (discs are avascular)

Discs lose hydration and height with age and prolonged sitting. Morning workouts after sleeping involve slightly taller, more hydrated discs — one reason trainers sometimes recommend lighter spinal loading early in the day.

SECTION 3: Joints — The Hinges of Movement

Classification of Joints

A joint (articulation) is where two or more bones meet. Joints are classified by structure and degree of movement:

Type	Example	Movement
Fibrous (synarthrosis)	Skull sutures	None
Cartilaginous (amphiarthrosis)	Pubic symphysis	Slight
Synovial (diarthrosis)	Knee, shoulder	Free

Synovial Joints — The Workhorse of Exercise

All major joints involved in gym exercises are synovial. They share common features: hyaline cartilage on articular surfaces, a joint capsule, synovial membrane producing synovial fluid, and associated ligaments.

Synovial fluid: A viscous lubricant that reduces friction, distributes load, and delivers nutrients to articular cartilage. Viscosity decreases with warming up — explaining why cold joints feel stiff and why warm-up is critical for injury prevention.

Synovial Joint Sub-Types in Fitness

Sub-Type	Shape	Exercise Example	Movements
Ball & socket	Spherical head + cup	Shoulder press	All planes + rotation
Hinge	Convex + concave	Bicep curl (elbow)	Flexion/Extension
Pivot	Rounded + ring	Forearm rotation	Rotation only
Condyloid	Oval + ellipsoid	Wrist (radiocarpal)	Flex/Ext + Abd/Add
Saddle	Reciprocal saddles	Thumb CMC	Biaxial
Plane (gliding)	Flat surfaces	Foot (intertarsal)	Gliding

Range of Motion (ROM)

ROM describes the arc of movement possible at a joint. It is constrained by:

- Bony architecture (e.g., hip socket depth — acetabular morphology varies 20–30° between individuals)
- Ligamentous tension (passive constraint at end-range)
- Muscle flexibility and fascial stiffness (active constraint through range)
- Joint capsule extensibility

Practical implication: Squatting depth is partly determined by hip socket anatomy. Attempting to force a squat deeper than one's bony architecture allows causes posterior pelvic tilt ("butt wink") — increasing lumbar flexion under load and disc stress. Programming should respect individual anatomy, not just mirror ideals.

SECTION 4: The Muscular System

Three Types of Muscle Tissue

Type	Control	Location	Characteristics
Skeletal	Voluntary	Attached to bones	Striated, fast & slow fibres, fatigable
Cardiac	Involuntary	Heart wall	Striated, rhythmic, highly aerobic
Smooth	Involuntary	Visceral organs	Non-striated, slow, sustained contraction

Skeletal Muscle Architecture

From macro to micro, skeletal muscle is organised in nested sheaths of connective tissue:

- Epimysium — outer fascial wrapping of the entire muscle belly
- Perimysium — surrounds fascicles (bundles of ~100 muscle fibres)
- Endomysium — wraps individual muscle fibres (cells)
- Sarcolemma — cell membrane of the muscle fibre
- Myofibrils — contractile units running the length of the fibre
- Sarcomeres — repeating units of myofibrils containing actin & myosin

The sarcomere is the fundamental unit of contraction. Under an electron microscope, alternating light (I-band, actin) and dark (A-band, myosin) bands create the characteristic "striped" appearance. During contraction, actin filaments slide past myosin — the sliding filament theory.

Muscle Fibre Types

Human skeletal muscle is a mixture of fibre types, genetically determined but modestly trainable. Understanding fibre types explains individual differences in athletic performance and guides programme design:

Property	Type I (Slow)	Type IIa (Fast-Ox)	Type IIx (Fast-Gly)
Colour	Red (myoglobin)	Red/Pink	White
Speed	Slow	Intermediate	Fast
Fatigue	Resistant	Intermediate	Rapid
Power output	Low	Moderate	High
Energy system	Aerobic (oxidative)	Mixed	Anaerobic (glycolytic)
Mitochondria	Many	Moderate	Few
Best for	Endurance	Middle distance	Sprint/Power

Fibre type distribution: The average untrained person has roughly 50% Type I and 50% Type II in most limb muscles, though this varies considerably between individuals and muscles (soleus is ~80% Type I; gastrocnemius ~50%). Elite marathon runners may be 80–90% Type I; sprinters the reverse.

Training Tip — Fibre Type Recruitment

- Endurance training (low load, high rep) preferentially recruits Type I fibres
- Heavy resistance training (>85% 1RM) recruits all fibre types including Type IIx
- Power training (explosive movement) converts some Type IIx to IIa (more sustainable)
- You cannot convert Type I to Type II — genetics sets the ceiling, training fills the potential

Upper Body Muscles

Chest (Pectoralis Major & Minor)

The pectoralis major is a large fan-shaped muscle with two heads:

- Clavicular head (upper chest) — originates from the clavicle; active in incline press
- Sternal/costal head (lower chest) — originates from sternum & ribs; primary flat bench muscle
- Action: horizontal adduction, flexion, and internal rotation of the humerus
- Synergists: anterior deltoid, triceps brachii (in pressing movements)

The pectoralis minor lies deep to pectoralis major, attaching from ribs 3–5 to the coracoid process of the scapula. It protracts and depresses the scapula — important for shoulder health and often tight in desk workers.

Back (Latissimus Dorsi, Trapezius, Rhomboids)

Muscle	Origin	Insertion	Primary Action	Key Exercises
Latissimus dorsi	Lower 6 thoracic vertebrae, iliac crest	Intertubercular groove of humerus	Shoulder extension, adduction, internal rotation	Pull-up, lat pulldown, row
Trapezius (upper)	Occipital bone, C1-C7	Clavicle, acromion	Scapular elevation & upward rotation	Shoulder shrug, overhead press
Trapezius (mid)	T1–T5	Spine of scapula	Scapular retraction	Rows, face pull
Trapezius (lower)	T6–T12	Root of scapular spine	Scapular depression	Y-raise, dip
Rhomboids	C7–T5	Medial scapular border	Scapular retraction & downward rotation	Rows, band pull-apart

Shoulder (Deltoids & Rotator Cuff)

The deltoid has three distinct heads that act as separate muscles:

- Anterior (front) — shoulder flexion, horizontal adduction; hit by pressing and front raises
- Lateral (side) — shoulder abduction; isolated by lateral raises (best at >30° abduction)
- Posterior (rear) — shoulder extension, horizontal abduction; engaged in rows and face pulls

Rotator Cuff (SITS muscles): Four small muscles surrounding the glenohumeral joint that stabilise the ball within the socket. They are crucial for injury prevention in overhead athletes and weightlifters:

Muscle	Primary Action
Supraspinatus	Shoulder abduction (initiation)

Infraspinatus	External rotation
Teres Minor	External rotation
Subscapularis	Internal rotation

Programming note: For every internal rotation movement (bench press, push-up, lat pulldown), programme a compensatory external rotation exercise (band ER, face pull) to prevent anterior shoulder dominance and impingement syndrome.

Arms (Biceps, Triceps, Forearms)

Muscle	Heads	Main Action	Best Exercise
Biceps brachii	2 (short + long)	Elbow flexion, supination	Barbell/dumbbell curl
Brachialis	1	Elbow flexion (strongest)	Hammer curl
Brachioradialis	1	Elbow flexion (neutral grip)	Reverse curl
Triceps brachii	3 (long/lat/med)	Elbow extension	Close-grip bench, dip, pushdown
Forearm flexors	Multiple	Wrist flexion, grip	Deadlift, carries

Lower Body Muscles

Quadriceps

The quadriceps femoris is a group of four muscles on the anterior thigh — the primary knee extensors and important hip flexors:

- Rectus femoris — only quad that crosses the hip; hip flexion + knee extension
- Vastus lateralis — outer thigh; often dominant, especially in women
- Vastus medialis — inner thigh; the teardrop shape visible in lean individuals; stabilises patella
- Vastus intermedius — deep, beneath rectus femoris; pure knee extension

The patellar tendon connects all four heads (via patella) to the tibial tuberosity. Patellar tendinopathy (jumper's knee) develops when repetitive tensile loads exceed the tendon's capacity — common in volleyball, basketball, and high-volume squatting.

Hamstrings & Glutes

Muscle	Origin	Insertion	Action
Biceps femoris (long)	Ischial tuberosity	Head of fibula	Hip extension + knee flexion
Biceps femoris (short)	Femur (linea aspera)	Head of fibula	Knee flexion only

Semitendinosus	Ischial tuberosity	Pes anserinus (tibia)	Hip extension + knee flexion + medial rotation
Semimembranosus	Ischial tuberosity	Medial condyle of tibia	Hip extension + knee flexion
Gluteus maximus	Ilium, sacrum, coccyx	Gluteal tuberosity, IT band	Hip extension + external rotation
Gluteus medius	Lateral ilium	Greater trochanter	Hip abduction + internal rotation
Gluteus minimus	Lateral ilium (lower)	Greater trochanter	Hip abduction + internal rotation

Why Glutes Matter Beyond Aesthetics

- Gluteus maximus is the largest muscle in the body — a primary power producer
- Weak glutes cause compensatory lumbar extension during hip-dominant movements
- Gluteus medius weakness = knee valgus (collapse inward) during squats and landing
- Hip thrust activates glute max ~30% more than squat; include both in programming
- Runners with weak glutes show higher rates of IT band syndrome and patellofemoral pain

Calves & Lower Leg

Muscle	Origin	Action	Best Exercise
Gastrocnemius	Posterior femoral condyles	Plantarflexion + knee flexion	Standing calf raise
Soleus	Posterior tibia/fibula	Plantarflexion (knee bent)	Seated calf raise
Tibialis anterior	Lateral tibia	Dorsiflexion + inversion	Toe raises
Peroneals	Fibula	Eversion + plantarflexion	Lateral ankle drills

SECTION 6: Core Anatomy

The Core — Beyond Abs

The "core" is not merely the rectus abdominis (the "six-pack" muscle). True core function involves a canister of muscles that stabilise the spine and pelvis during virtually every movement:

Layer	Muscle	Primary Role
Anterior	Rectus abdominis	Spinal flexion, intra-abdominal pressure
Lateral	External oblique	Rotation, lateral flexion
Lateral	Internal oblique	Rotation (opposite), lateral flexion
Deep	Transversus abdominis	Circumferential stabilisation, IAP

Posterior	Erector spinae	Spinal extension, posture
Posterior	Multifidus	Segmental spinal stabilisation
Floor	Pelvic floor	Intra-abdominal pressure, pelvic organ support
Ceiling	Diaphragm	Breathing, intra-abdominal pressure

Intra-abdominal pressure (IAP) is the hydrostatic pressure within the abdominal cavity generated when the diaphragm descends, pelvic floor contracts, and transversus abdominis co-activates. This creates a rigid cylinder protecting the spine during heavy loading — the physiological basis for the Valsalva manoeuvre in powerlifting.

SECTION 7: Tendons, Ligaments & Connective Tissue

Tendons vs Ligaments

TENDONS	LIGAMENTS
Connect muscle to bone	Connect bone to bone
Transmit muscle force	Constrain joint movement
Primarily Type I collagen	Type I + Type III collagen mix
Moderate blood supply	Poor blood supply (slow healing)
Respond well to progressive loading	Limited capacity for hypertrophy
Can store elastic energy	Primarily passive restraints

Tendon adaptation to training: Unlike muscle, tendon takes 6–12 months to meaningfully strengthen. This creates a "weak link" window when muscle strength outpaces tendon adaptation — a common cause of overuse injuries in people who progress too rapidly. This is why progressive overload must be gradual.

Fascia — The Body's Connective Web

Fascia is a continuous web of connective tissue enveloping every muscle, organ, nerve, and blood vessel. Far from passive wrapping, fascia:

- Transmits force between muscles across joints (myofascial slings)
- Contains proprioceptors — contributing to body position sense
- Becomes stiffer ("densified") with dehydration, immobility, and chronic stress
- Responds to slow, sustained stretching (yin yoga, foam rolling) and hydration

SECTION 8: Introduction to Muscle Hypertrophy

What Makes Muscles Grow?

Hypertrophy (muscle growth) occurs when muscle protein synthesis (MPS) exceeds muscle protein breakdown (MPB) over time. Three primary mechanisms drive hypertrophy:

Mechanism	Description	Training Method
Mechanical tension	Force on actin-myosin cross-bridges — the dominant driver	Heavy compounds, full ROM, controlled eccentrics
Metabolic stress	Lactate accumulation, swelling, cell signalling	Moderate weights, high reps, short rest, pump work
Muscle damage	Eccentric-induced micro-tears triggering satellite cell activity	Novel exercises, slow eccentrics, full stretch

Research increasingly shows mechanical tension is the primary driver. Metabolic stress and muscle damage are secondary pathways. This explains why heavy, progressive training (even if reps are low) produces substantial hypertrophy when volume is adequate.

Satellite cells are muscle stem cells that reside beneath the basal lamina. Following mechanical damage or metabolic stress, they activate, proliferate, and fuse with existing fibres, donating their nuclei. More nuclei = greater synthetic capacity = larger muscle cell.

mTORC1 (mechanistic target of rapamycin complex 1) is the primary anabolic signalling hub. Resistance training and leucine (from dietary protein) both activate mTORC1, which upregulates ribosomal biogenesis and protein synthesis. This molecular basis explains why post-workout protein consumption amplifies training adaptations.

Module 1 Key Takeaways

- The skeleton provides structure, leverage, and protection — bone is a living, adaptable tissue
- Synovial joints enable the wide range of motion required for exercise; anatomy varies between individuals
- Three muscle fibre types (I, IIa, IIx) differ in speed, fatigue resistance, and metabolic pathway
- Major muscle groups have distinct origins, insertions, and actions — knowing them prevents injury and improves targeting
- Hypertrophy requires mechanical tension as the primary stimulus, aided by adequate protein intake
- Tendons adapt more slowly than muscles — the primary reason for gradual progression rules

Quick Review Questions

1. What is the difference between compact and cancellous bone, and why does this matter for exercise?

2. Name the four rotator cuff muscles and explain their collective function.

3. How do Type I and Type IIx muscle fibres differ in terms of energy system use and fatigue?

4. Explain the sliding filament theory of muscle contraction.

5. Why is the transversus abdominis considered the most important "core" muscle for spinal stability?